SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2023	**Date:**	25/08/26

Code of Conduct

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Jahnavi

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules.
   Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.
   An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships

- ACTIVATE(Roaming, Customer) → Action: ACTIVATE, Object: Roaming, Agent: Customer.
- DEACTIVATE(CallerTune, Customer) → Action: DEACTIVATE, Object: CallerTune, Agent: Customer.
- QUERY(DataBalance, Customer) → Action: QUERY, Object: DataBalance, Agent: Customer.
- ACTIVATE(5GService, Customer) → Action: ACTIVATE, Object: 5GService, Agent: Customer.

In every case the predicate structure follows the pattern ACTION(ENTITY, AGENT), where the verb of the query is mapped to the predicate name and the requested service becomes the argument entity.

2. Queries with Semantic Interpretation Errors

Q2 and Q4 contain interpretation errors. In Q2 the user intent "Deactivate Caller Tune" was wrongly predicted as "Activate Caller Tune" — the semantic analyzer correctly captured the object (CallerTune) but reversed the action, producing an opposite meaning. In Q4 the user intent "Enable 5G Service" was predicted as "Activate 5G Service"; although ENABLE and ACTIVATE are near-synonyms and the outcome is functionally similar, this still reflects an imprecise sense mapping since the analyzer did not distinguish between the two related but distinct action verbs. Q1 and Q3 show a correct match between actual and predicted intent.

3. Effect of Meaning Representation on Chatbot Accuracy

The action predicate is the single most decision-critical element of the representation, because the downstream system executes the operation named by the predicate. A wrong action predicate, as in Q2, causes the chatbot to perform the exact opposite of what the customer asked for (activating instead of deactivating a paid service), which directly damages customer trust and can create billing disputes. Even a near-miss such as Q4 shows that loosely defined predicate vocabularies (ENABLE vs ACTIVATE, CHECK vs QUERY) can silently reduce precision metrics even when the practical effect is harmless. This demonstrates that decision accuracy in a chatbot is only as good as the fidelity of its semantic parser.

4. Recommendations for Telecom Chatbot Systems

- Standardize a controlled predicate vocabulary (ACTIVATE, DEACTIVATE, QUERY, MODIFY) and map all verb synonyms to it via a lexicon, rather than allowing the parser to invent near-duplicate predicates.

- Introduce negation-aware parsing so that antonymous verbs such as "activate" and "deactivate" are never confused, e.g. by explicit polarity tagging on the predicate.
- Add a confirmation step before executing state-changing actions (activate/deactivate), where the bot restates the parsed intent back to the user for validation.
- Continuously retrain the intent classifier on misclassified logs (such as Q2 and Q4) to improve verb-sense disambiguation over time.

Q2. First-Order Predicate Calculus for Smart Manufacturing

1. FOPC Representation of Production Data

- Active(M1), Active(M2), Active(M4)
- Maintenance(M3)
- Rule 1: $\forall x$ [Active(x) $\rightarrow$ Producing(x)]
- Rule 2: $\forall x \forall y$ [Produces(x,y) $\wedge$ Active(x) $\rightarrow$ Available(y)]
- Rule 3: $\forall x$ [Maintenance(x) $\rightarrow$ $\neg$Producing(x)]

2. Inference — Which Products Are Available

Applying Rule 1 to the active machines: Active(M1), Active(M2) and Active(M4) each entail Producing(M1), Producing(M2) and Producing(M4). By Rule 2, for any product y such that a producing, active machine x Produces(x,y), Available(y) is inferred. Therefore every product manufactured by M1, M2, or M4 is currently Available. M3 is in Maintenance, so by Rule 3, $\neg$Producing(M3) holds, and Rule 2 cannot be triggered for M3 — any product that depends solely on M3 is not marked Available.

3. Is Gear Production Affected by Maintenance?

If Gear is produced only by M3 — i.e., Produces(M3, Gear) — then since Maintenance(M3) $\rightarrow$ $\neg$Producing(M3) by Rule 3, Producing(M3) is false, so Rule 2's antecedent Produces(M3,Gear) $\wedge$ Active(M3) cannot be satisfied (Active(M3) is false), and Available(Gear) cannot be derived. Hence Gear production is directly affected and halted while M3 remains under maintenance. However, if an alternate active machine (e.g., M1, M2 or M4) also satisfies Produces(x, Gear), then Available(Gear) can still be inferred through that machine, meaning Gear supply would be unaffected at the product level even though M3's individual contribution is halted.

4. Effectiveness of Predicate Logic in Industrial Decision Support

First-order predicate calculus gives the plant a precise, machine-checkable way to encode operational states and automatically derive downstream consequences (availability, shortages) without manual tracking. Its strengths are unambiguous semantics, easy rule auditing, and straightforward extension with new rules. Its limitations are that it handles only crisp, binary states — it cannot represent partial or probabilistic availability (e.g., a machine running at reduced capacity), and every new real-world nuance requires an explicit new rule. For richer decision support, predicate logic is best combined with probabilistic or fuzzy reasoning layers.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

1. Correct Word Sense for Each Query

- "Apple accessories" → Technology Brand sense (confirmed by the click on "iPhone Charger").
- "Mouse wireless" → Computer Device sense (confirmed by the click on "Bluetooth Mouse").
- "Java tutorial" → Programming Language sense (confirmed by the click on "Coding Lessons").
- "Python course" → Programming Language sense (confirmed by the click on "Software Development Training").

2. Semantic Cues Used for Disambiguation

Two main cue types resolve the ambiguity. First, co-occurring context words within the query itself act as local disambiguators — "accessories" pairs naturally with the Technology-Brand sense of Apple, "wireless" pairs with the Computer-Device sense of Mouse, and "tutorial"/"course" pair with the Programming-Language senses of Java and Python. Second, behavioral/click-through evidence from the search logs provides an external, data-driven confirmation of sense choice — the actual product or content the user selected after searching reveals which sense they intended, independent of the query text alone.

3. Impact of Incorrect Sense Selection

If the engine picked the wrong sense — e.g., returning fruit-related results for "Apple accessories" or island-travel content for "Java tutorial" — the user would see irrelevant results, leading to lower click-through rate, poor conversion, increased query reformulation, and reduced user trust in the platform's search relevance. At scale, systematic sense errors on high-volume ambiguous terms can materially reduce revenue and degrade the personalization/recommendation models trained on that click data.

4. Industrial-Scale WSD Strategy

- Use contextual embeddings (e.g., transformer-based encoders) that assign sense-specific vector representations based on surrounding query words rather than static, single-meaning word vectors.
- Continuously mine click-through and dwell-time logs as weak supervision signals to retrain and validate sense-selection models.
- Incorporate user-level personalization/history (e.g., past purchases in electronics vs. groceries) to bias sense selection toward the individual's likely intent.
- Maintain a domain-specific sense inventory/knowledge base (product taxonomy) so the disambiguation model is anchored to actual catalog categories rather than generic dictionary senses.

Q4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

1. Syntactic Structures Contributing to Semantic Role Assignment

All four sentences follow a Subject–Verb–Object (SVO) syntactic pattern, and this structure is the primary cue used to map grammatical positions onto semantic roles. Generally, the syntactic subject of an action verb is mapped to the Agent role (the entity performing the action), the syntactic object is mapped to the Recipient/Theme (the entity affected by the action), and any additional noun phrase denoting a means is mapped to the Instrument role. For example, in "Doctor prescribed medicine to patient," the subject "Doctor" becomes Agent, the direct object "medicine" becomes Instrument (the means by which treatment is delivered), and the indirect object "patient" becomes Recipient.

2. Are the Assigned Semantic Roles Appropriate?

Mostly yes, but with one inconsistency. Doctor = Agent and Patient = Recipient are correctly assigned for the prescribing sentence. Medicine = Instrument is reasonable in "Doctor prescribed medicine to patient" and is confirmed again in "Medicine reduced blood pressure," where Medicine actually behaves as the Agent/Causer of the reduction rather than a mere instrument, since it is the subject performing the action in that sentence. Patient reporting a headache should assign Patient = Agent/Experiencer (the one experiencing/reporting the symptom) and Headache = Symptom (Theme), which matches the given table. Overall the roles are broadly appropriate, but Medicine's role shifts between Instrument and Agent/Causer depending on sentence context, which the flat role table does not capture — a sign that role assignment needs to be done per-sentence rather than as a single global entity-to-role mapping.

3. Potential Errors From Incorrect Parsing

If the parser mis-tags constituents — e.g., treating "to patient" as a locative rather than a recipient phrase, or misidentifying the head noun in a complex noun phrase — the resulting role assignment could reverse Agent and Recipient, causing the system to record that the patient administered medicine to the doctor. In clinical NLP this is especially dangerous: swapping Agent/Recipient in medication or dosage sentences, or misclassifying a symptom as an instrument, can corrupt structured records used for clinical decision-making, drug interaction checks, or automated alerts, potentially leading to patient-safety risks.

4. Recommendations to Improve Syntax-Driven Semantic Analysis in Healthcare

- Use dependency parsing (not just shallow SVO tagging) to correctly resolve prepositional attachments (e.g., "to patient") into the right semantic role.
- Fine-tune parsers and role labelers on domain-specific, clinically annotated corpora rather than generic-English treebanks, since medical sentence structures and terminology differ from everyday text.

- Assign roles per-sentence/per-predicate context (frame semantics) instead of one static role per entity, so an entity such as Medicine can be correctly labeled Instrument in one sentence and Agent/Causer in another.
- Add a human-in-the-loop clinical validation step for high-risk extractions (e.g., dosage, medication administration) before they are used in automated decision-support pipelines.